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National Football League Players Have Higher Rates of Knee Extensor Mechanism Tears During Short and Normal Rest Weeks versus Long Rest Weeks

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Conflict of Interest: The authors declare that they have no conflict of interest.

Abstract:
Background: Lower extremity injuries, particularly knee injuries, in National Football League (NFL) players are common. Extensor mechanism ruptures often require season-ending surgery. Research on NFL injury risk factors, including time between games, is growing, but studies on rest periods and knee injuries have shown mixed results. No prior research has specifically examined extensor mechanism tears in relation to short, normal, or long rest periods between games.

Methods: An online search was conducted to identify NFL players who tore their quadriceps tendon or patellar tendon between the 2009-2023 seasons. Only regular season injuries after week 1 were included. The date of injury and prior game date were collected in order to calculate the length of rest. Next, each injury was categorized as a short (<7 days), normal (7 days), or long (>7 days) week injury. Other factors, such as age, football position, and playing surface (natural grass or artificial turf) were also collected.

Results: A total of 58 players met inclusion criteria. There was no difference in extensor mechanism injury rates between short weeks and normal weeks. Players were 4.7 times more likely to tear their extensor mechanism during normal weeks than long weeks ($p = 0.0039$) and 3.7 times more likely to tear their extensor mechanism during short weeks than long weeks ($p = 0.0385$). Regarding playing surface, extensor mechanism tears were more likely to occur on artificial turf than natural grass ($p = 0.0324$).

Conclusion: During the 2009-2023 NFL seasons, players were found to have 4.7 times higher rates of knee extensor mechanism tears during short (< 7 days) rest weeks and 3.7 times higher rates during normal (7 days) rest weeks between games, when compared to long (>7 days) rest weeks. There was no difference in knee extensor mechanism injury rates between the short and normal rest weeks.

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Abstract

Background: Lower extremity injuries, particularly knee injuries, in National Football League (NFL) players are common. Extensor mechanism ruptures often require season-ending surgery. These ruptures, involving structures like the quadriceps and patellar tendons, are caused by powerful movements that overload the knee. Research on NFL injury risk factors, including time between games, is growing, but studies on rest periods and knee injuries have shown mixed results and are limited in scope. No prior research has specifically examined extensor mechanism tears in relation to short, normal, or long rest periods between games.

Methods: Using publicly accessible resources (NFL injury reports, Pro-Football-Reference.com, and ESPN.com), an online search was conducted to identify NFL players who tore their quadriceps tendon or patellar tendon between the 2009-2023 seasons. Only regular season injuries after week 1 were included. The date of injury and prior game date were collected in order to calculate the length of rest. Next, each injury was categorized as a short (<7 days), normal (7 days), or long (>7 days) week injury. Other factors, such as age, football position, and playing surface (natural grass or artificial turf) were also collected.

Results: A total of 58 players met inclusion criteria. There was no difference in extensor mechanism injury rates between short weeks and normal weeks. Players were 4.7 times more likely to tear their extensor mechanism during normal weeks than long weeks ($p = 0.0039$) and 3.7 times more likely to tear their extensor mechanism during short weeks than long weeks ($p = 0.0385$). Regarding playing surface, extensor mechanism tears were more likely to occur on artificial turf than natural grass ($p = 0.0324$).

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Conclusion: During the 2009-2023 NFL seasons, players were found to have 4.7 times higher rates of knee extensor mechanism tears during short (< 7 days) rest weeks and 3.7 times higher rates during normal (7 days) rest weeks between games, when compared to long (>7 days) rest weeks. There was no difference in knee extensor mechanism injury rates between the short and normal rest weeks.

Level of Evidence: Level III

Clinical Relevance: Shorter than average time between games in the NFL does not seem to increase risk of knee extensor mechanism injuries, but longer than average time between games seems to protect against these injuries.

Introduction

Among all injuries in the National Football League (NFL), lower extremity injuries are the most common, specifically injuries of the knee.[1] One knee injury a player may experience is a rupture of the extensor mechanism. The knee extensor mechanism, composed of the quadriceps muscle, quadriceps tendon, medial/lateral patellar retinaculum, patella, patellar tendon, and tibial tubercle, plays a crucial role in active knee extension and resistance of knee flexion. NFL players are at risk of extensor mechanism ruptures, specifically at the quadriceps tendon or patellar tendon, due to powerful movements resulting in tensile overloads on the extensor mechanism.[3] These injuries often lead to season-ending surgery.[4–6]

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Injury risk factors in NFL players is an expanding area of research, as rule changes can be implemented to increase player safety and limit injuries. Each year, the NFL competition committee releases new rule change proposals, with the goal of injury prevention.[7] Notable changes have occurred in order to prevent lower extremity injuries, prevent concussions, and protect players in vulnerable game situations.[7,8] One potential injury risk factor is time between games, which few studies have explored. One study found no association between knee injuries and extrinsic risk factors, including time between games.[9] However, another study found players are at a lower overall injury risk during short weeks (less than 7 days between games).[10] These studies were limited in that they only observed two seasons and three seasons respectively, and they did not distinguish between different types of knee injuries. A recent study specifically explored the risk of anterior cruciate ligament (ACL) tears in NFL players based on rest week length.[11] It was found that deviation from the normal 7-day NFL week increased the risk of ACL tears, as short and long rest weeks had higher risk of tears compared to normal rest weeks.[11]

To date, there have been no studies analyzing the risk of extensor mechanism tears based on short (less than 7 days), normal (7 days), or long (more than 7 days) rest periods during an NFL season. Thus, this study aimed to determine the risk of knee extensor mechanism tears among NFL players based on short, normal, or long rest weeks. We hypothesized that short and long NFL rest weeks will increase the risk of extensor mechanism tears compared to normal rest weeks.

Methods

Data Collection

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This was a retrospective, observational study in which an online search was conducted to compile a list of NFL players who tore their quadriceps tendon or patellar tendon between the 2009-2023 seasons using publicly accessible data. Online searches were performed on each publicly accessible resource. Publicly accessible resources used to create the dataset included NFL team injury reports, ESPN.com, and Pro-Football-Reference.com.[12–14] Keyword searches included “knee injury,” “quadriceps tendon,” and “patellar tendon” to extract necessary injury data. Player demographic data, position, and age at time of injury were collected from Pro-Football-Reference.com.[13] Date of injury, game week number, and prior game date were collected from NFL injury reports, Pro-Football-Reference.com, and other publicly available sources. Playing surface type (natural grass or artificial turf) was determined from NFL.com and official NFL team websites.[12]

Each injury was categorized as a short, normal, or long week injury based on the time between the date of injury and the previous game date the player participated in. A short week injury was defined as an injury that occurred less than 7 days after the previous NFL game. A normal week injury was defined as an injury that occurred 7 days after the previous NFL game. A long week injury was defined as an injury that occurred greater than 7 days after the previous NFL game.

Only regular season injuries between the 2009-2023 seasons were included, while preseason injuries, postseason injuries, and practice injuries were excluded. Week 1 injuries were also excluded due to inconsistencies in rest following the preseason.

Statistical Analysis

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Descriptive statistics were calculated to report means, ranges, and percentages. Football position type grouping was as follows: defensive secondary, linebacker, offensive lineman, defensive linemen, quarterback, wide receiver, special teams, tight ends, and running backs. Quadriceps and patellar tendon tear risk during short, normal, and longer weeks were analyzed based on position group, week length in days, player age, body mass index (BMI), and playing surface. Fisher's exact or chi square analysis was used to examine relationships for categorical variables, and one-way ANOVA was used to examine relationships between continuous variables. The level of significance was established at a 2-sided alpha level of $p<0.05$. All statistical analyses were performed utilizing Python (version 3.12; Python Software Foundation).

Results

During regular season games between the 2009-2010 and 2022-2023 seasons, 58 NFL players tore their extensor mechanism (EM), with 20 being quadriceps tendon (QT) tears, and 38 being patellar tendon (PT) tears. In regard to week length, 8 EM tears (4 QT, 4 PT) occurred during short weeks, 47 EM tears (16 QT, 31 PT) occurred during normal weeks, and 3 EM tears (0 QT, 3 PT) occurred during long weeks (Table 1). In this same time period, 4887 regular season games were played, excluding preseason, week 1, and postseason games. A total of 701 games were played during short weeks, 3219 were played during normal weeks, and 967 were played during long weeks.

Rates of extensor mechanism tears were 0.011 for short weeks, 0.015 for normal weeks, and 0.003 for long weeks. After statistical analysis, NFL players were found to be 4.7 times more likely to tear their extensor mechanism during normal rest weeks than long rest weeks ($p = 0.00386$). Short rest weeks were associated with 3.7 times more injury risk than long rest weeks

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(p = 0.03846). There was no difference between short rest weeks and normal rest weeks (p = 0.5157).

Regarding player position, 26 players were offensive (5 wide receivers, 16 offensive linemen, 4 tight ends, 1 running back), 30 players were defensive (15 defensive linemen, 7 linebackers, 8 defensive secondary), and 2 players were special teams (1 kicker, 1 punt returner). Average BMI was 33 ± 4.2 kg/m2 and average age at injury was 27 ± 3 years (Table 2).

For overall extensor mechanism tears, there was no association between injury rates and player position (p = 0.8618), age (p = 0.4497), or BMI (p = 0.7310). Among QT tears, there was no association between injury rates and player position (p = 0.7481), age (p = 0.8373), or BMI (p=0.9294). Finally, among PT tears, there was no association between injury rates and player position (p = 0.9770), age (p = 0.3860), or BMI (p=0.9055).

Regarding playing surface, 26 EM injuries occurred on grass (8 QT, 18 PT), while 32 occurred on artificial turf (12 QT, 20 PT). This revealed an association between overall extensor mechanism injuries and artificial turf playing surface (p = 0.0324). There was no association between playing surface and QT (p = 0.3044) or PT (p = 0.0801) injuries when isolated (Table 3).

Discussion

The primary finding from this investigation was that NFL players were 4.7 times more likely to tear their extensor mechanism during normal rest weeks than long rest weeks, and short rest weeks were associated with 3.7 times more injury risk than long rest weeks. We found no difference in knee extensor mechanism injury rates between short rest weeks and normal rest

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weeks in NFL players. Therefore, our hypothesis suggesting that short and long rest weeks would have increased risk of injury compared to normal rest weeks, was rejected.

It is well established that fatigue from overuse increases the risk of muscular injuries and tendinous injuries.[15–17] However, the effect of variable rest times between NFL games on injury rates remains controversial. Multiple studies have reported no increased risk of injury during shorter game weeks,[9,18] and some even report less injuries during Thursday night NFL games.[10,19] These results do not align with our findings that short and normal weeks have an increased risk of extensor mechanism injury, when compared to long rest weeks. The current study differs from previous studies in methodology. For example, Lawrence et al.[9] categorized injuries into broad categories, such as knee, ankle, hamstring, shoulder, and concussion injuries. Additionally, they only included injuries during the 2012-2013 and 2013-2014 seasons. The current study investigated a more specific injury pattern, being patellar tendon and quadriceps tendon tears, which are significant injuries often requiring surgery. The authors wanted to investigate the relationship between rest time and a severe injury, as this has large consequences, such as time off, decreased performance, and even retirement for some players.[5,20,21] The current study also includes a longer time period than the aforementioned studies.

Our finding that normal and short rest weeks have higher risk of extensor mechanism injury compared to longer rest weeks has plausible physiological explanations. Reduced recovery time can lead to increased tendon stiffness in the knee,[22] and mechanical loading without proper recovery can cause biochemical and gene expression responses from mechanosensitive cells, leading to extracellular matrix degradation.[22] Due to extracellular matrix degradation, the tissue capacity is decreased, and a subsequent mechanical overload may lead to tissue failure and injury.[22]

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When comparing short and normal rest weeks, we found no significant difference in extensor mechanism tear rates. This may have multiple potential explanations. First, muscular and tendinous recovery may be similarly decreased at both time points, resulting in similar injury rates between short and normal rest weeks. Furthermore, tendons have slower tissue renewal rates compared to muscles,[23] which may lead to similar tendon stiffness between short and normal rest periods. Compared to American football, many soccer leagues have more frequent matches.[24] One study found that match congestion (number of days separating two matches) in soccer players was not associated with overall injury rate.[24] However, muscular injury rates were lower after at least 6 days of rest between matches as compared to 3 days or less. Interestingly, there was no difference in muscular injuries between 3 or less days of rest, 4 days of rest, or 5 days of rest.[24] This pattern is similar to the findings of our study, which suggests that short rest periods are equally detrimental when compared to adequate rest.

Sharma et al. found that Thursday NFL games had no difference in risk of injury, but games on Monday night did have lower rates of injuries.[18] The current study supports these findings, as longer rest periods were shown to have the lowest rates of extensor mechanism injury. However, the current study differs because groups were stratified based on days between games, not the day of the week itself. Sharma et al. also reported that defensive linemen, linebackers, skilled running positions, and wide receivers had decreased injury rates on Monday night games. This contrasts with the current findings of no association between positional grouping and extensor mechanism injury rates.

Many NFL players have been critical of shortened rest weeks, as the traditional, normal rest week allows for decreased practice intensity and greater opportunity for recovery prior to the next game.[25] In contrast, when preparing for a Thursday night game, this process is unable to

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happen due to the shortened rest week, leading to a perceived increased risk of injury.[25,26] In addition, mental fatigue and stress may have detrimental effects on injury risk. In soccer, studies have found that mental fatigue significantly impairs decision-making ability in match play situations,[27] as well as physical performance.[28] This suggests that a mentally fatigued player may encounter more high-risk scenarios which may lead to injury. On the other hand, when teams have a game on Monday night, they have a longer rest week prior to that game, which allows for increased rest. Bye weeks, or entire weeks off, have the same effect. Some argue that when an NFL team has a Thursday night game, it leads to a "mini-bye week," because the following game on the next week's Sunday would lead to 9 days of rest between the two games.[29] Either way, long periods of rest seem to be preferred by NFL players,[29] but a Thursday night game will always result in a shortened rest period leading up to the game. The current study supports that shortened rest does not increase the risk of injury, compared to normal rest weeks. Furthermore, the resulting long period of rest afterwards may be beneficial, as long rest weeks have the lowest risk of injury to the extensor mechanism.

Limitations

A limitation of this study is that the data is collected from publicly accessible resources, such as NFL injury reports. This may result in information being left out, reporting biases, and incorrect information. NFL teams are required to publish injury reports each week, which must include all minor and major injuries. Due to quadriceps tendon and patellar tendon tears being major, season-ending injuries that often require surgery, there is expected to be minimal unreported injuries for the current study. Another limitation of the current study is that the length of rest in days may not be truly representative of the rest that each individual NFL player

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receives. The differences in practice intensity and rehabilitation may affect the frequency of injuries. Another limitation of the study is its small sample size; however, this was expected due to the investigation of a specific injury pattern with inclusion criteria of only regular season game injuries.

Conclusions

During the 2009-2023 NFL seasons, players were found to have 4.7 times higher rates of knee extensor mechanism tears during short (< 7 days) rest weeks and 3.7 times higher rates during normal (7 days) rest weeks between games, when compared to long (>7 days) rest weeks. There was no difference in knee extensor mechanism injury rates between the short and normal rest weeks. Furthermore, the resulting long period of rest after a short rest week may be beneficial, as long rest weeks have the lowest risk of injury to the extensor mechanism.

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Table 1: Quadriceps Tendon, Patellar Tendon, and All Extensor Mechanism Injuries Characterized by Short, Normal, and Long Weeks

Short Week (<7 days between NFL games)	Normal Week (7 days between NFL games)	Long Week (>7 days between NFL games)	Total
Quadriceps Tendon Injuries			
4	16	0	20
Patellar Tendon Injuries			
4	31	3	38
All Extensor Mechanism Injuries			
8	47	3	58

Key: NFL-National Football League

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Table 2: National Football League Positions, Number of Injuries, Age, and BMI Characterized by Extensor Mechanism Injury

Position	Number of Injuries	Age* (sd)	BMI^ (sd)
Quadriceps Tendon Injury			
Defensive Linemen	8	29.4 (2.9)	34.8 (2.4)
Offensive Linemen	5	27.4 (2.9)	37.5 (2.0)
Linebacker	4	28.3 (3.7)	31.2 (1.3)
Defensive Back	1	28	30
Tight End	1	30	32
Kicker	1	26	28
Patellar Tendon Injury			
Offensive Linemen	11	27.5 (2.3)	36.3 (1.1)
Defensive Linemen	7	27.0 (3.3)	35.4 (3.8)
Defensive Back	7	26.7 (2.6)	27.6 (1.3)
Wide Receiver	5	26.4 (1.0)	27.1 (0.4)
Tight End	3	26.0 (2.9)	30.6 (1.1)
Linebacker	3	26.0 (1.6)	30.3 (1.3)
Running Back	2	24 (1.0)	27.5 (1.3)
All Extensor Mechanism Injuries			
Offensive Linemen	16	27.4 (2.5)	36.7 (1.5)
Defensive Linemen	15	28.3 (3.4)	35. 1 (3.2)
Defensive Back	8	26.9 (2.5)	27.9 (1.5)
Linebacker	7	27.3 (3.2)	30.8 (1.4)
Wide Receiver	5	26.4 (1.0)	27.1 (0.4)
Tight End	4	27.0 (3.1)	30.9 (1.1)
Running Back	2	24 (1.0)	27.5 (1.3)
Kicker	1	26	28

Key: BMI-Body Mass Index; sd-Standard Deviation; *-reported in years; ^-reported in kg/m^2

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Table 3: Extensor Mechanism Injuries Characterized by Playing Surface in National Football League Players

	Playing Surface		
	Natural Grass	Artificial Turf	p Value
Quadriceps Tendon	8	12	0.3044
Patellar Tendon	18	20	0.0801
All Extensor Mechanism	26	32	**0.0324***

Key: *-denotes statistically significant value

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